MIME 497 CAPSTONE I
DESIGN PROPOSAL

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MIME 497
SUMOBOT TEAM ORANGE

INTRODUCTION

Over the course of winter and spring term, Sumobot team orange will be working on a capstone project to present in June at the end of the school year. The intent of the capstone project is to showcase the skillset which has been built over an undergraduate engineering career at Oregon State University. The purpose of this project is to develop a fully functional robotic system capable of competing in the school's Sumobot competition. The Sumobot competition is one where teams create robots to face off against each other. A team will win the competition if their robot can push their opponents' out of a sumo wrestling-like ring. In this project, team orange will build a maximum 3kg robot, which will fit in a 20cm x 20cm square, and will be controlled by a remote controller. The teams have been given winter and spring term of 2026, and will compete and present their projects at the 2026 Oregon State University Engineering Expo.

PROJECT SCOPE

This project is needed by both the college of engineering and the students themselves to showcase the skills which the college has developed in the students. This project will exercise the various disciplines taught in the Oregon State Engineering curriculum. The project will be presented to Oregon State faculty, and the scope of this project includes having a working robot to compete by June. This project has the potential to be marketed as a commodity for consumers but that is not in the scope of this project. The focus of this particular design challenge is to demonstrate engineering skills, adhere to a ruleset and customer requirements, and outperform the competition.

RESEARCH & PREVIOUS APPROACHES

We have conducted research about the sumobot competition including researching its rules, as well as previous approaches that have been taken in the design process. This research gives that a robot may expand after the match has started. This means that robots may have the ability to transform to dimensions which extend the 20cm x 20cm dimension limit. Competing robots may also be built with no height limit. This research gives information about the dimensions of the arena, start locations, as well as the arena material. This information will be implemented in the design of the robot. Previous iterations of the sumobot project include one which was designed to be very tall and expanding outward after match start. This particular design utilized a vacuum seal base with a swinging arm to extend beyond the 20cm x 20cm dimension square to knock opponents off the arena. Teams have utilized height and a "wall structure" to push opposing robots out of the circle. Teams have also used a magnet which creates downforce to the sumo plate. Another approach is using a track movement system like a tank, rather than wheels for extra traction along the plate.

CONSTRAINTS, RISK & ASSUMPTIONS

There are constraints associated with the project including budget, timeframe, and ruleset. The budget is $1000, offered by the college of engineering. Teams may source more money for the project through sponsors; however, team orange will assume to only utilize the budget provided by the college. The time constraint is a hard deadline June 1. For competing in the sumobot competition at the engineering expo in June at Oregon State University, the robot must be fully functional by then. This project is constrained by various rules, such as the 20cm x 20cm dimension limit and the 3kg weight limit. The risk associated with this project includes robot breaking, parts being delayed, cost risk due to redesigning the robot, and schedule risk such as delayed testing or designing.

STAKEHOLDERS

Stakeholders are groups of people who are affected by the project, because of their involvement, or because of the project's outcome. In this case, the stakeholders of this competition are, first, the Oregon State College of Engineering students. Direct group members of team orange are involved in the project directly and will be affected by the project process regarding learning outcomes, intellectual property, and the projects effects on career opportunities and outcomes. The next Capstone class at the University are also stakeholders. They will be able to research this project and use the iterations of this project for future inspiration. Second, the Instructor for this course, which will be evaluating the capstone projects. Third, Oregon State University is a stakeholder, which sets the required learning outcomes for the curriculum and specifically for this course and provides the sumobot team with a budget to abide by. Lastly any companies which we source components from such as parts suppliers will be involved in this project indirectly, making them a stakeholder.

TEAM APPROACH

Team orange will use a remote control for the robot. The goal is to create a robot which maximizes the space and weight allowed within the ruleset. There is a set of constraints which include weight limit, maximum dimension, a budget set by the school, and a deadline of June. Team orange has decided to design the robot based on a tank vehicle, utilizing a track drivetrain system. Team orange is currently designing an offensive mechanism which takes advantage of the expandable ruleset after start, and will utilize this research in designing the robot.

DESIGN DECISION

In planning for the design, team orange has developed customer requirements to decide based on the sumobot competition ruleset. Included in these customer requirements is:

- The robot must be able to produce 8 lbs. of force
- Robot can move along the x and y plane easily
- Robot must be around 3kg +/- 0.1kg
- Robot must fit within a 20cm by 20cm box
- Robot top speed of 2 feet per second
- Robot must withstand a force of 9lbs without moving
- Robot must decelerate at a rate of at least 6ft/s^2
- Robot must use RC control for operation
- Outsourced robot parts must be receivable within 1 week
- Robot build must include one form of offensive capability

Based on these customer requirements, the team generated hand drawn ideas independently to use in a Pugh decision matrix. The matrix included 6 ideas and a datum (Fig.1):

Ideas	Datumn	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5	Idea 6
Criteria							
Robot can prooduct atleat 8lbs of force		+	-	+	+	+	-
Robot can move along the x and y plane easily		-	+	+	-	-	-
Robot must be around 3kg +/- 0.1kg		+	+	+	+	+	+
Robot must fit within a 20cm by 20cm box		+	+	+	+	+	+
Robot top speed of 2 feet per second		-	+	-	-	-	-
Robot must withstand a force of 9lbs without moving		+	-	+	+	+	-
Robot must decelerate of at least a rate of 6ft/s^2		-	-	-	+	-	-
Robot must use RC control for operation		+	+	+	+	+	+
Robot part must be accesible within one day of shipping		-	+	+	-	-	-
Robot build must include one form of offensive capability		-	-	-	+	-	+
Sum of positives	5	5	6	7	7	5	4
Sum of negatives	5	5	4	3	3	5	6

Our group came to to conclusion that ideas 3 and 4 would fit our engineering specifications the best: low frame, prioritizing a high torque drive system, and having one offensive capability. From this we will start to to create low fidelity prototypes to design a robot that fit these concepts to further progress this project.

Figure 1

The matrix gave 3 and 4 as inspiration for the robot. The goal is to combine traits between the two ideas listed here:

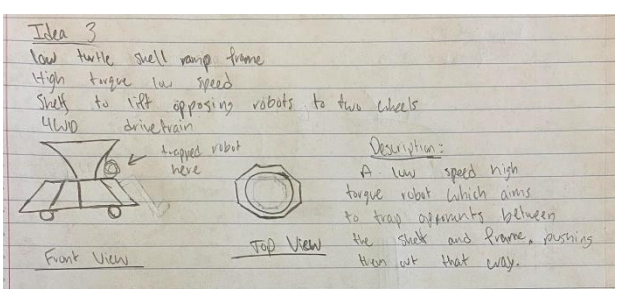

Figure 2: Idea 3: A turtle shell frame design with a high torque low speed drive system (Fig.2)

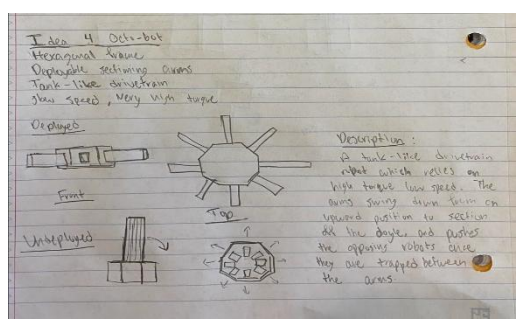

Figure 3: Idea 4: A hexagonal frame design with a tank drive system (Fig.3)

DESIGN PROCESS

Team orange will use a tank-like drivetrain, with a ramped frame, and a high torque low speed drive system. The robot will have 4 high torque motors. SolidWorks 3D CAD Software is being used to design the robot and create different iterations of the product. Here is the current 3D model:

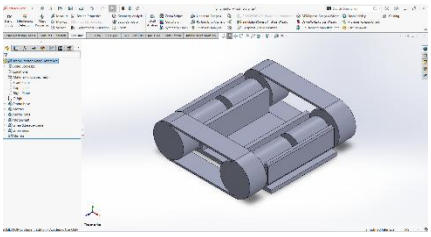

Figure 2

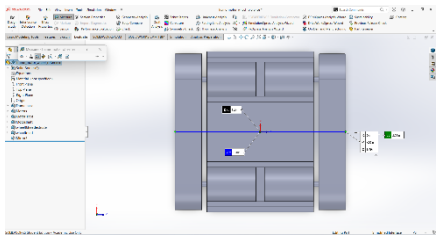

Figure 5

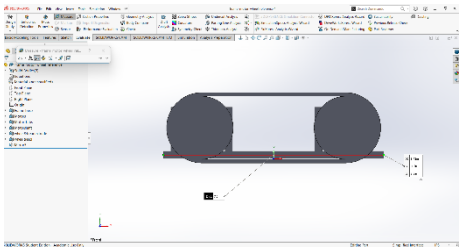

Figure 6

These images show the isometric, top, and side view of the CAD software progress. The robot treads are shown, as well as bounding bodies which represent the motor dimensions. There is a center space for the power supply and the controller (Arduino). The created frame is dimensioned to fit within the 20 x 20 cm box as per the customer requirements. This is the baseline idea for the drivetrain system for our robot.

LOW FIDELITY PROTOTYPE

A low fidelity prototype has been created to visualize how the design choices will fit together in the real world. The low fidelity prototype is shown below (Fig.7, Fig.8), created out of cardboard geometry with accurate dimensioning:

Figure 7

Figure 8

CONCLUSION

In conclusion, this project is aimed at creating a thorough understanding of engineering in the real world, by using various disciplines to create a robot to compete amongst others in the sumobot competition. Team orange will create a fully functioning robot by June ready to compete, which is within the scope of our project. Upon research from past sumobot iterations, team orange will use a remote controller for operating the robot, maximizing the weight limit, and creating the largest robot allowed within the dimensions. Team orange will utilize the entire budget given by the school and order supplies to create the robot. Team orange will use SolidWorks 3D CAD Software to design the sumobot and save part files. Further prototyping will be done using 3D printing. The prototypes will increase in accuracy starting from the low fidelity prototype made from cardboard material.

Sources

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